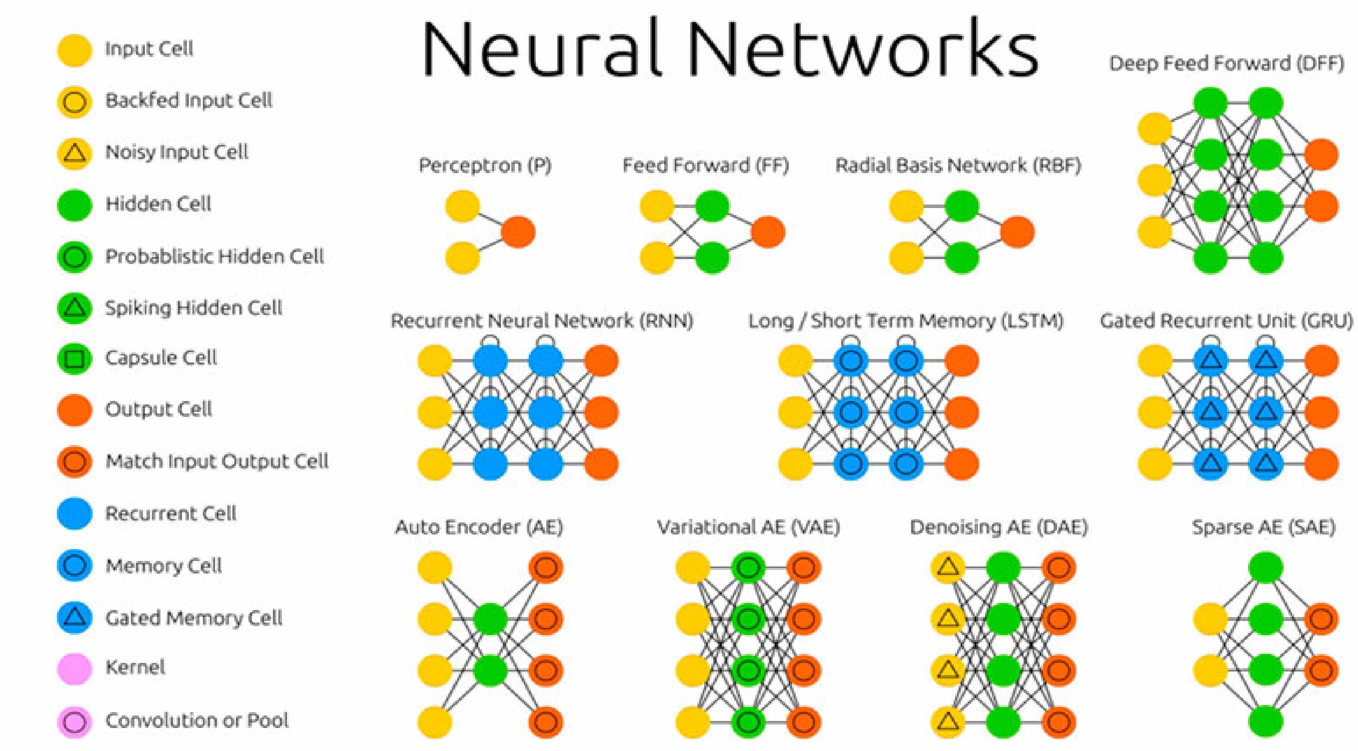
Neural networks

16/12/2024

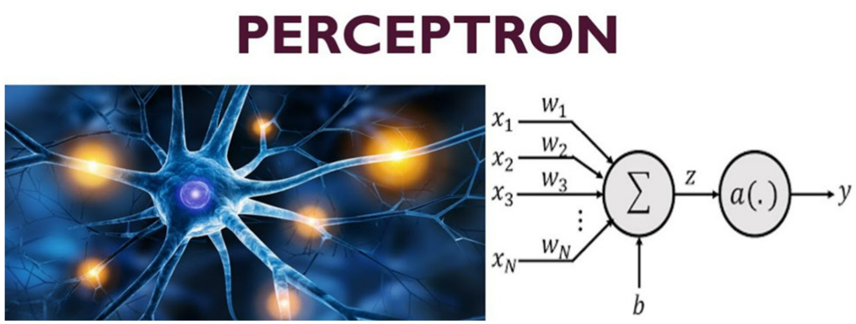

Neural Networks
Input Cell
Backfed Input Cell
Noisy Input Cell
Hidden Cell
Probablistic Hidden Cell
Spiking Hidden Cell
Capsule Cell
Output Cell
Match Input Output Cell
Recurrent Cell
Memory Cell
Gated Memory Cell
Kernel
Convolution or Pool
Perceptron (P)
Feed Forward (FF)
Radial Basis Network (RBF)
Deep Feed Forward (DFF)
Recurrent Neural Network (RNN)
Long / Short Term Memory (LSTM)
Gated Recurrent Unit (GRU)
Auto Encoder (AE)
Variational AE (VAE)
Denoising AE (DAE)
Sparse AE (SAE)

Perceptron

Perceptron

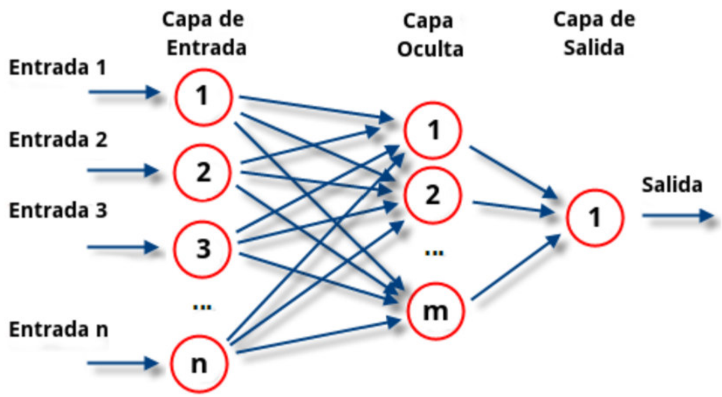
Basic unit of a neural network.

Used for classification and regression.



Multilayer Perceptron (MLP)

Feedforward Neural Network, that connects neurons in different layers.



Feed Forward Neural Networks

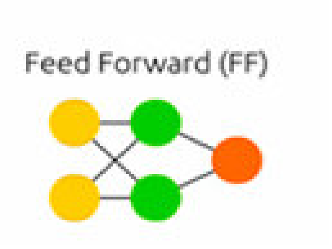
Perceptrons and radial basis function networks transform patterns from input to output.

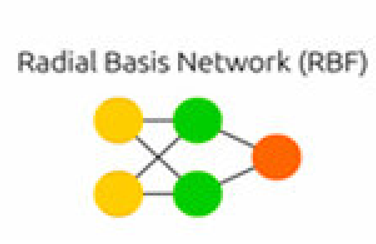
They have layers that consist of either input, hidden or output nodes.

Nodes are connected between adjacent layers. Fully connected, when every neuron from one layer to every neuron in another layer.

The minimal network has two input cells and one output cell that can be used to model logic gates, for example.

Backpropagation is a common learning algorithm where the network is shown pairs input and expected output, and the strength of the connections between nodes is updated based on the model's success in predicting.

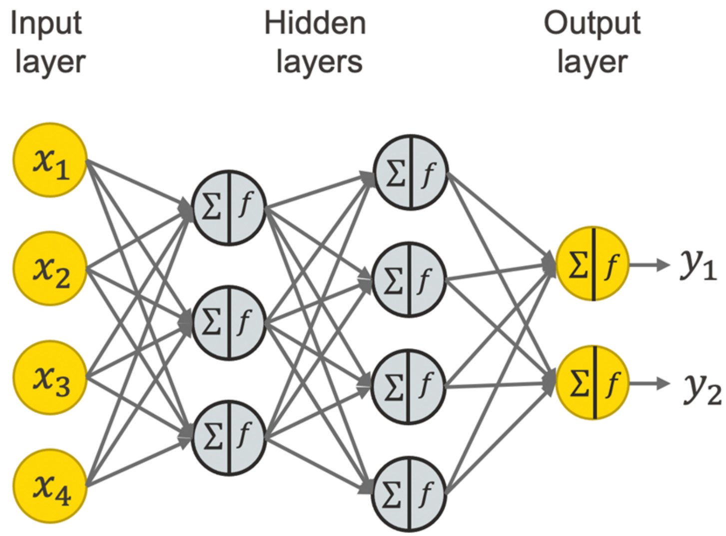




Deep Learning

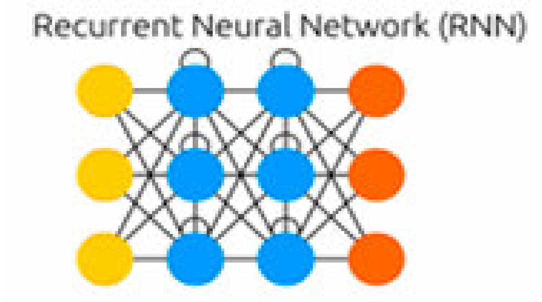
Deep neural networks are capable of learning complex relationships of the data.

- Convolutional neural networks.
- Recurrent neural networks.



Recurrent neural networks

Recurrent networks are feedforward networks with connections within layers. Therefore, they are not stateless and the timing and order in which input is structured matters. This allows recurrent networks to find structure in time.



Training these networks may yield vanishing (or exploding) gradients, where, depending on the activation functions used, information gets lost (or amplified) over time, similar to how very deep feedforward networks can lose information in depth.

Long Short-Term Memory

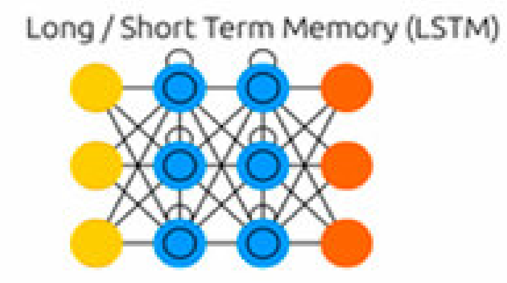
LSTMs provide a resolution for the vanishing and exploding gradient problems, by introducing gates and explicitly defined memory cells.

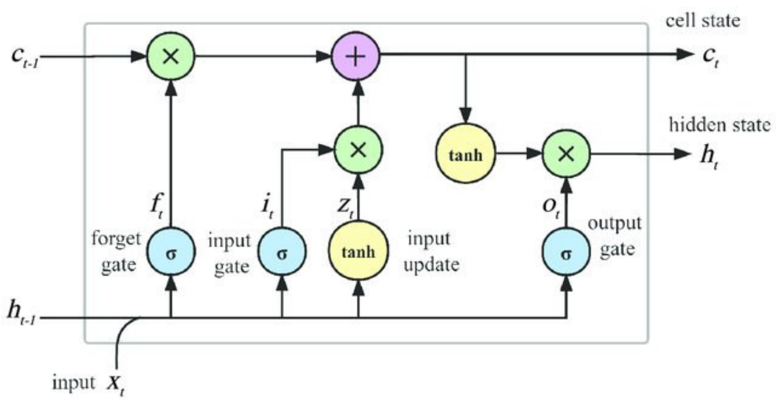
Each node has a memory cell and three gates: Input, output and forget.

The function of these gates is to protect the loss of information by stopping or allowing it to flow.

The input gate determines how much of the information from the previous layer is stored in the cell. The output gate determines what the next layer gets to know about the state of this cell. The forget gate prevents new information from being ignored.

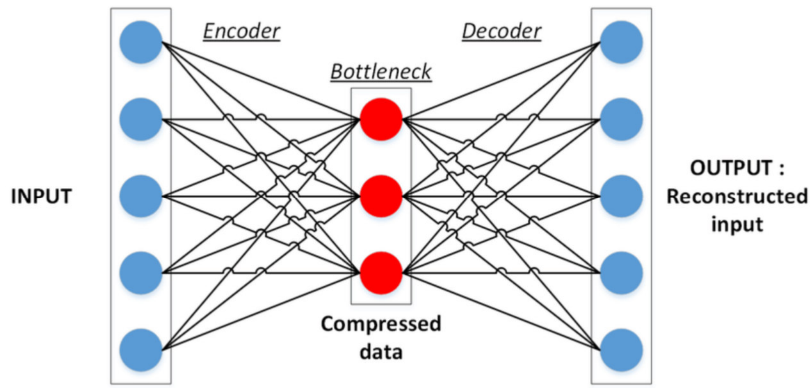
Variants: Gated recurrent units.





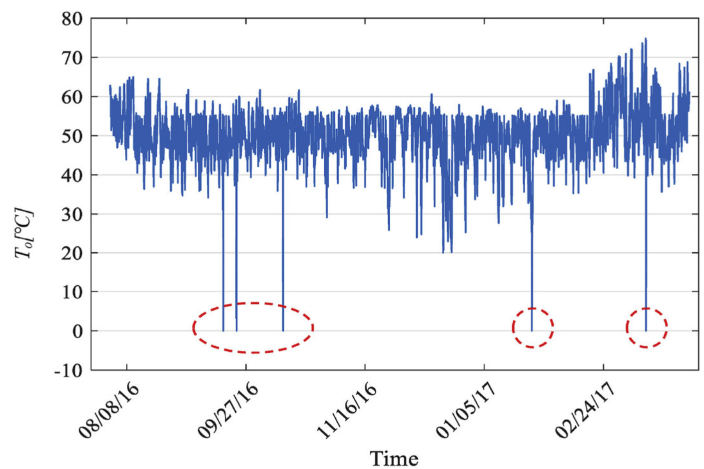
Autoencoders

Autoencoders compress (encode) and regenerate (decode) information by transforming it through a smaller hidden layer with symmetrical surrounding layers. The resemblance between input and output can be used as a measure of success for the compression.



Anomaly Detection

- Train with normal data.
- Evaluate with new data. Data which is significant different from the training data has a high reconstruction error.



Autoencoders

Image denoising

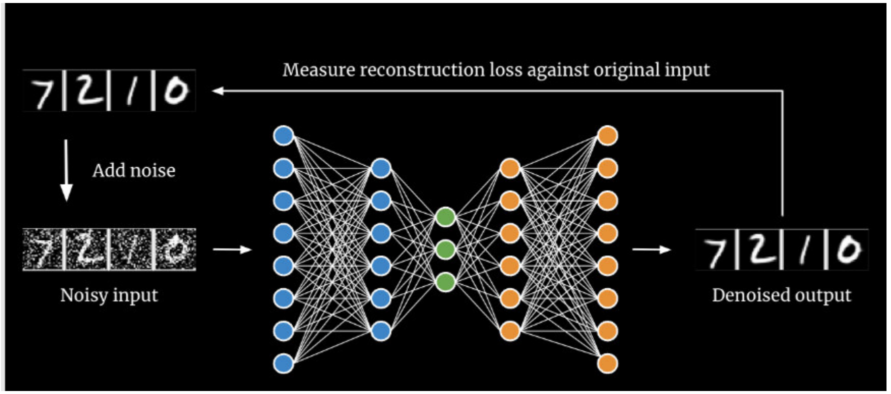
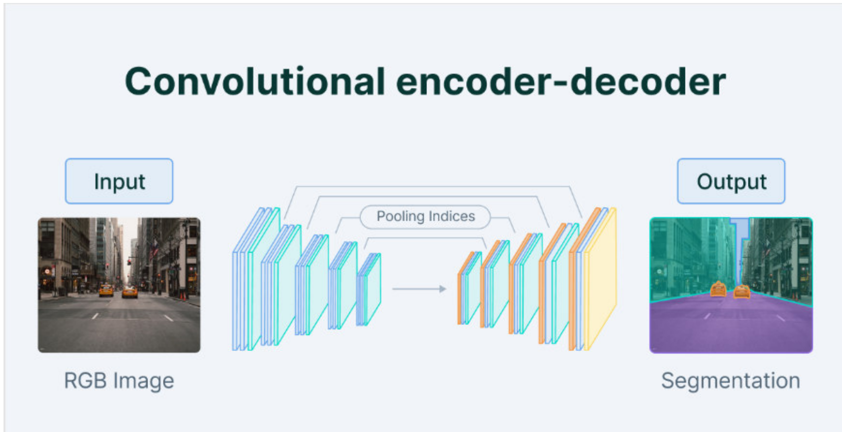


Image segmentation



Variational Autoencoders

Variational autoencoders share a similar architecture as autoencoder, but instead, learn an approximate probability distribution of the input patterns grounded in Bayesian inference and modeling causal relations.

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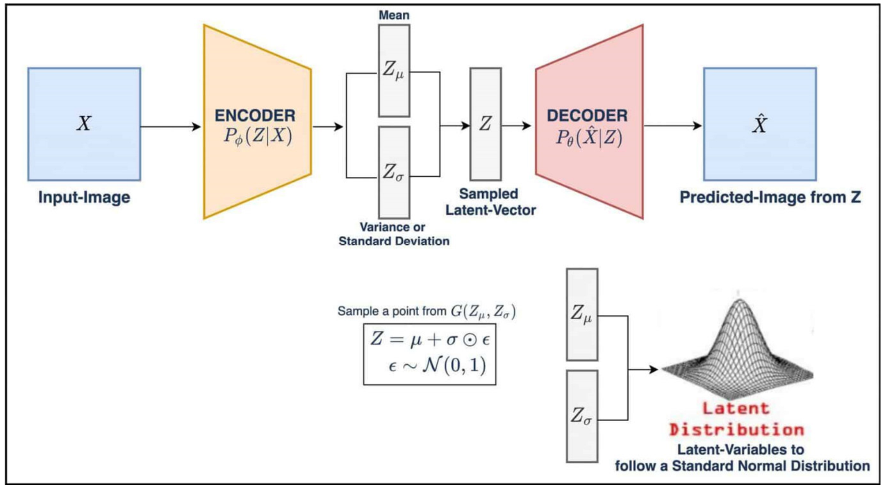
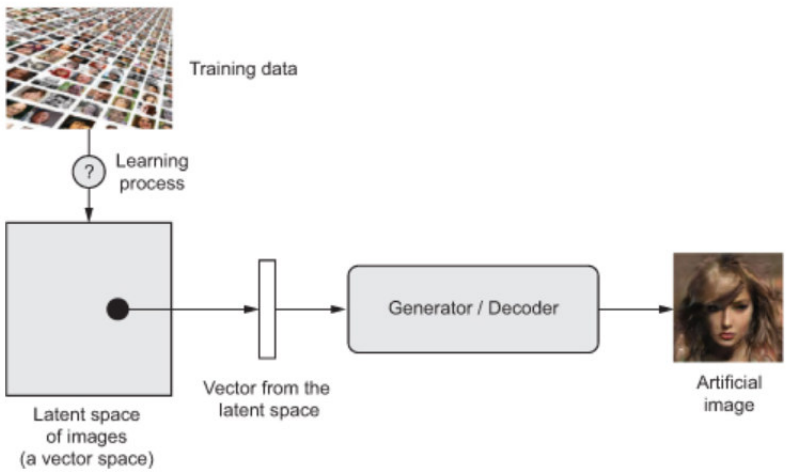


VAE: Generating new images

- Latent space is well structured.
- Specific directions encode a meaningful axis of variation in the data.



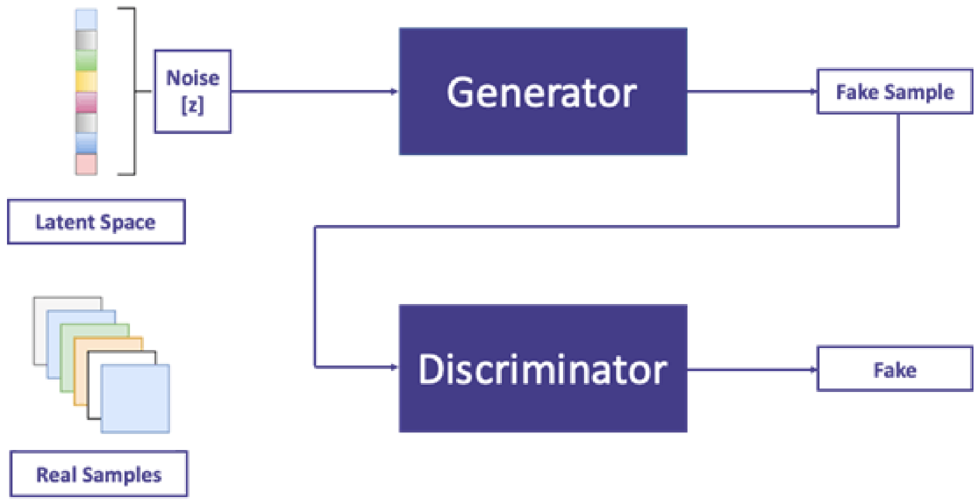
Generative adversarial networks

Consist of two networks:

- Generator: generates data.
- Discriminator: predicts whether the data have been generated or not.

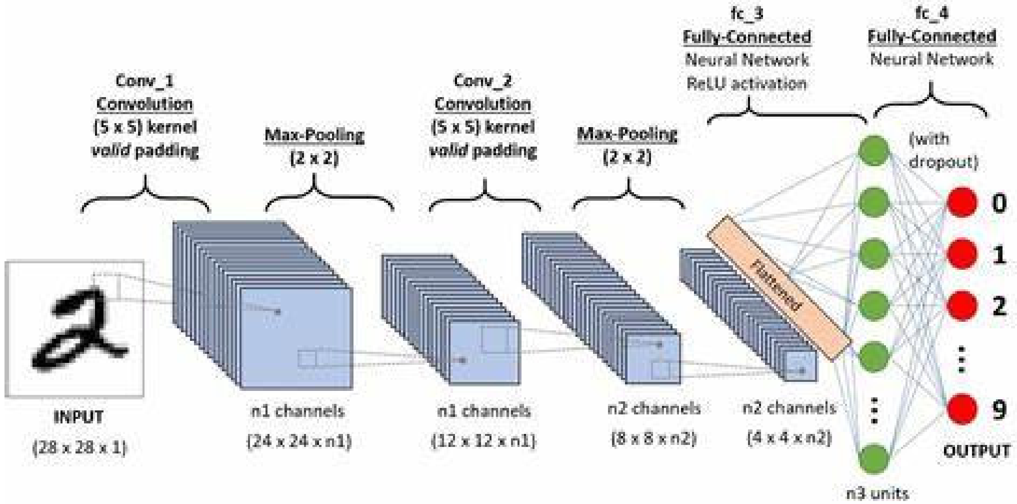
The predictive success of the discriminator is used as an error gradient for the generator.

This setup aims for the discriminator to get better at distinguishing real data from generated data, while the generator learns to become less predictable.



Convolutional neural networks

Convolutional networks are deep learning architectures that typically contain convolutional and pooling layers, used for approximate scanning of patterns that are often spatially correlated.



State of the art for object recognition

- YOLO (You Only Look Once), Faster R-CNN
- AlexNet, Inception-v3 , VGGNet, ResNet (Residual Network)

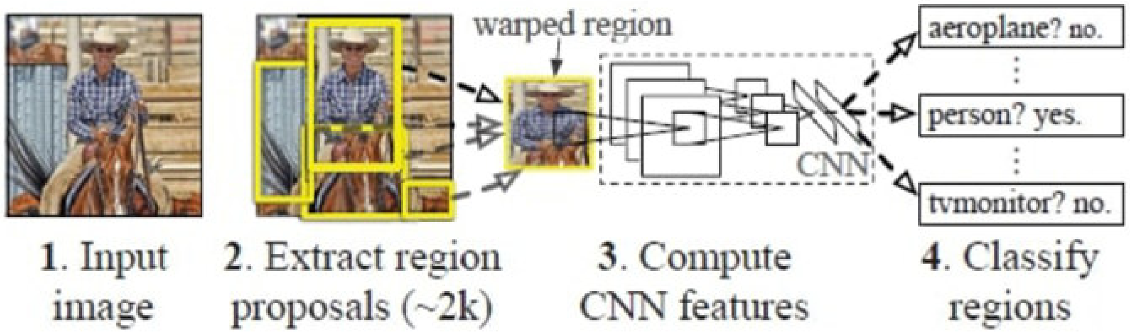
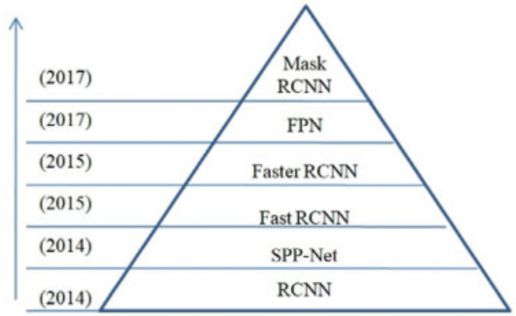
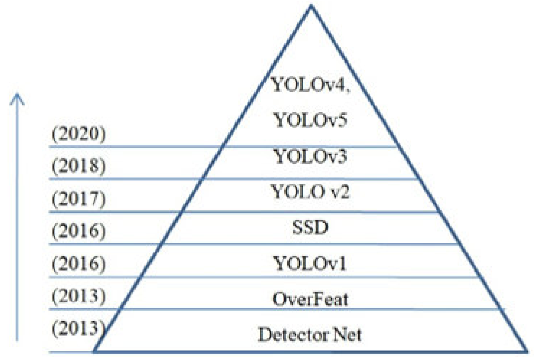


Fig. 2. Architecture of RCNN [25].



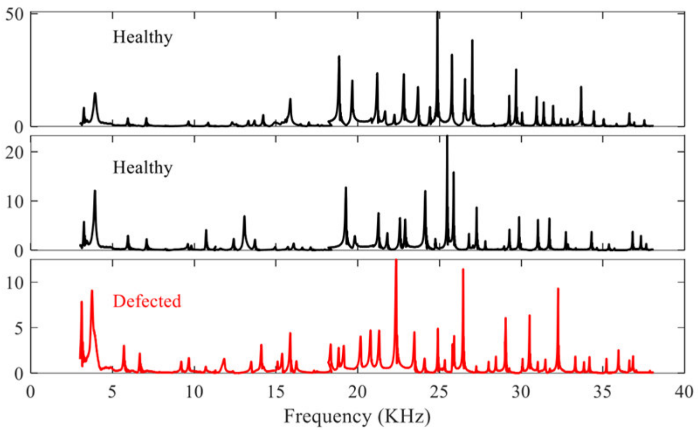
(a) Two-Stage Object detector



(b) One-Stage Object detector

Deep learning machine failure detection

Sensorial data



Training a CNN network

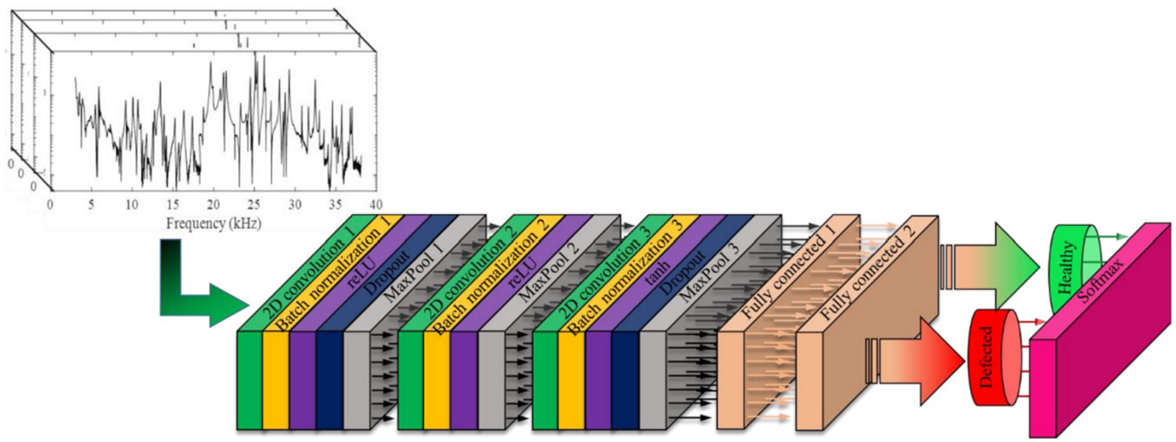
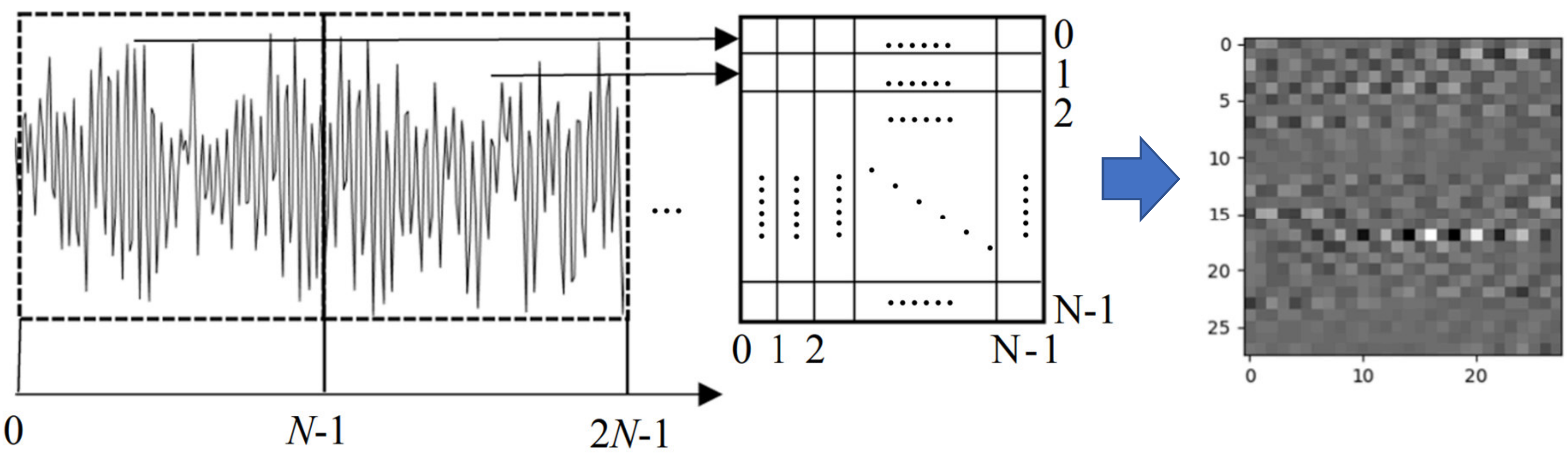


Figure 3. The proposed CNN structure

CNN identification of failures

Conversion of vibration signal to images.



CNN failure identification

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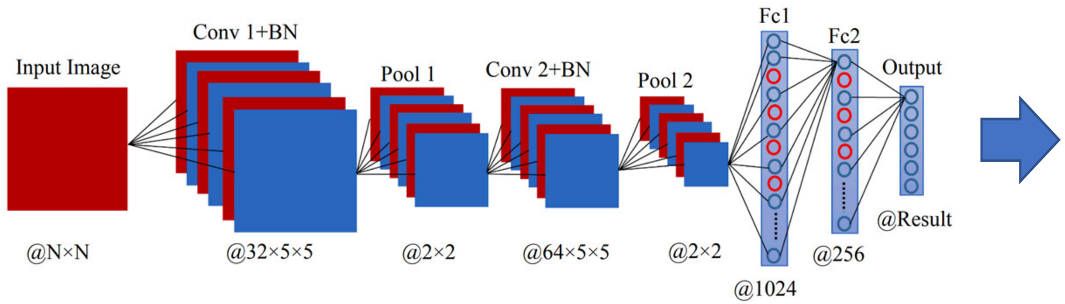
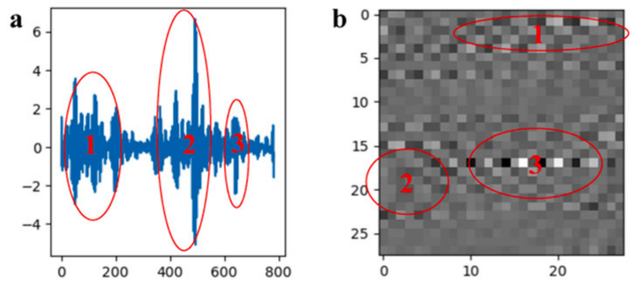
Definition of different type of failures.





Activation Maps

Visualization of activations in different layers of the CNN.

Discovery of characteristics patterns for the recognition of objects/failures.

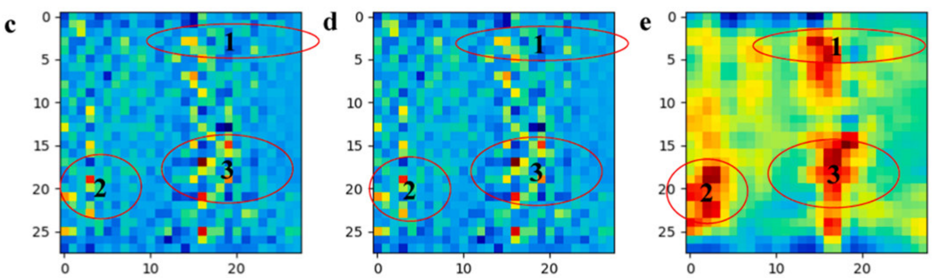


Figure 4. (**a**) Original image; (**b**) Class Activation Map; (**c**) results of Grad-CAM; (**d**) results of Grad-CAM++; and (**e**) GC-MRCNN. Various original images and its results of Grad-CAM, Grad-CAM++, and the proposed algorithm (GC-MRCNN).